Notes: Garmaise & Moskowitz (RFS, 2003)
Informal Financial Networks: Theory and Evidence

Contents

1 Big picture

- **Question.** Can non-financial intermediaries - brokers, lawyers, suppliers, accountants, consultants - improve borrower access to finance through informal relationships with lenders?
- **Setting.** The paper studies U.S. commercial real estate transactions from COMPS.com over 1992-1999. The empirical intermediary is the property broker; the lender is usually a bank providing new mortgage finance.
- **Theory.** A service intermediary repeatedly sends clients to lenders. A lender can reciprocate by expediting loan evaluations for that intermediary's clients. This repeated relationship can support cooperation even though the broker is not itself a financial intermediary.
- **Central mechanism.** Borrowers transact infrequently and cannot promise future business to a lender. Brokers transact repeatedly and can direct future clients to a lender. That repeated flow of business gives brokers bargaining/relationship value.
- **Main empirical result.** Controlling for endogenous broker selection, brokered deals are much more likely to obtain new bank financing.
- **Network evidence.** Brokers concentrate their bank business among a small number of banks, and brokers with stronger or longer relationships have larger effects on access to bank loans.
- **Loyalty evidence.** Loyal broker-bank relationships increase the probability of bank financing; disloyal broker-bank behavior reduces it.
- **Alternative theories.** Certification, advice, and broker selection cannot explain the full set of results. In particular, brokered deals do not receive higher prices after instrumentation, and broker-bank concentration/loyalty is hard to explain with simple certification.
- **Economic takeaway.** Even in a developed U.S. capital market, informal financial networks can affect who gets bank credit. Access to finance is partly determined by repeated relationships that sit outside formal loan contracts.

2 Data and measurement

2.1 Commercial real estate transaction data

- The sample comes from COMPS.com, a provider of U.S. commercial real estate sale data.
- The period is January 1, 1992 to March 30, 1999.
- The raw data contain 36,678 commercial real estate transactions; 22,642 meet the paper's data requirements.
- The data cover eleven states plus the District of Columbia.
- Required information includes sale price, financing details, identity/location of principals, property location, and broker activity.

Interpretation: The dataset is unusually rich because it observes the financing structure of individual real estate transactions and the presence of intermediating brokers.

2.2 Property and market characteristics

Key property controls include:

- property age;
- development status;

- capitalization rate, defined as net operating income divided by sale price;
- property type: apartment, vacant land, or commercial/industrial;
- personal crime and property crime near the property;
- whether the property is in a city center.

Interpretation: These controls are important because broker use and financing may both be correlated with property quality, local risk, and property type.

2.3 Market participant variables

- COMPS reports the locations of buyers and sellers.
- The authors compute buyer and seller distance from the property using latitude and longitude.
- Out-of-state status is also observed.
- Broker presence equals one when a professional broker participates in the transaction.
- The paper also controls for cases where the buyer or seller is itself a broker.

Interpretation: Distance and out-of-state status proxy for information frictions. Remote buyers or sellers may need brokers more, and lenders may view them differently.

2.4 Financing variables

The paper distinguishes three main financing forms:

- **New mortgage finance:** a new bank loan originated for the transaction.
- **Vendor-to-buyer finance (VTB):** seller financing.
- **Assumed mortgage finance:** buyer assumes an existing mortgage.

For each financing type, the paper studies both:

- the **frequency** of use, usually with a binary dependent variable;
- the **magnitude** when used, usually as financing amount divided by sale price.

Interpretation: The distinction between frequency and magnitude matters because brokers may help deals obtain bank financing without necessarily increasing the size of the loan conditional on receiving one.

2.5 Broker-bank relationship measures

The empirical network measures include:

- **Broker concentration among banks:** whether a broker sends many deals to a few banks.
- **Bank Herfindahl / bank share:** concentration measures for each broker's distribution of transactions across banks.
- **Broker-bank longevity:** length of the prior relationship between a broker and bank.
- **Broker loyalty:** whether the broker's business with a bank changes proportionally with the bank's overall market share.

- **Broker herding:** whether brokered deals cluster disproportionately at particular banks relative to what would be expected by chance.

Interpretation: The key idea is that a pure certification story predicts brokers improve loan access, but not necessarily that they repeatedly channel clients to the same banks. Relationship-specific concentration and loyalty are direct network evidence.

3 Theory and empirical specifications

3.1 Basic informal-network model

There are buyers, sellers, lenders, and service intermediaries. Buyers have limited wealth and need financing of amount I. Buyers are either good or bad. Lenders can evaluate applicants in two ways:

- **Standard evaluation:** cheaper effort cost e_0, but slow; only a fraction of good opportunities are completed.

- **Expedited evaluation:** higher effort cost $e_1 > e_0$, but fast; all good opportunities can be exploited.

The paper assumes:

$$\text{lenders do not approve without evaluation,} \tag{1}$$
$$(p-q)v > e_1 - e_0, \tag{2}$$
$$\frac{(p-q)v}{2} < e_1 - e_0. \tag{3}$$

Interpretation: Expedited evaluation is socially efficient, but not privately worthwhile for a lender in a one-shot transaction. Repeated broker-lender interaction makes it privately sustainable.

3.2 Cooperative dyadic relationships

For a sufficiently high discount factor, there is an equilibrium in which a long-run broker repeatedly directs clients to one lender, and the lender expedites evaluation for that broker's clients. If either party deviates, the relationship breaks down.

Interpretation: The broker creates relationship capital by promising future client flow. The lender reciprocates by providing a costly service - faster evaluation - that increases completed transactions and broker fees.

3.3 Predicted probability of bank financing

The paper derives that brokered transactions are more likely to receive bank finance. The key inequality is:

$$1 > \frac{2q\varepsilon}{q(2\varepsilon - \max\{\varepsilon - f, 0\}) + p\max\{\varepsilon - f, 0\}} > \frac{2q}{q+p}. \tag{4}$$

Interpretation: The left term corresponds to the highest bank-finance probability for clients of cooperating brokers; the right term is the lower probability for nonbrokered completed transactions. The result motivates Prediction 1.

3.4 Main empirical specification: new bank financing

The baseline second-stage probability model is:

$$\Pr(NewMortgage_i = 1) = \Phi\left(\beta \widehat{Broker}_i + X_i'\gamma + \delta_s\right), \tag{1}$$

where X_i includes property, buyer, seller, location, crime, and price controls, and δ_s are state dummies.

The loan-size model is:

$$\frac{NewMortgage_i}{Price_i} = \beta\widehat{Broker}_i + X_i'\gamma + \delta_s + u_i, \tag{2}$$

estimated with the Cragg truncated regression model conditional on positive new mortgage finance.

Interpretation: The first equation asks whether brokers improve access to bank credit. The second asks whether brokers change the amount of bank credit conditional on getting a loan.

3.5 Instrumenting broker presence

Broker use is endogenous, so the paper instruments broker presence with local market variables:

- population radius;
- personal crime rate;
- local capitalization-rate dispersion;
- fraction of brokered deals done by national brokers nearby;
- local broker Herfindahl index.

The first stage is:

$$Broker_i = Z_i'\pi + X_i'\eta + \delta_s + v_i. \tag{3}$$

Interpretation: The instruments are intended to capture local broker-market structure and broker availability rather than the unobserved financing need of the transaction.

3.6 Relationship specifications

To test whether broker-bank relationships matter, the paper estimates models of the form:

$$\Pr(NewMortgage_i = 1) = \Phi\left(\beta\widehat{Broker}_i + \theta(\widehat{Broker}_i \times Relationship_i) + X_i'\gamma + \delta_s\right). \tag{4}$$

$Relationship_i$ is proxied by concentration, bank share, broker-bank longevity, or loyalty.

Interpretation: A positive θ means the effect of a broker is larger when the broker has stronger relationship capital with the bank.

4 Identification and interpretation

4.1 What identifies the main broker effect

- The raw correlation between broker use and bank finance is not sufficient, because brokers may be selected into difficult or easy deals.
- The paper therefore uses instruments for broker presence.
- Regressions include property, buyer, seller, crime, price, and state controls.
- Coefficients are reported as Fama-MacBeth time-series averages across the sample period.

Interpretation: The empirical design is not a randomized experiment. Its credibility depends on the claim that local broker-market instruments affect financing primarily through broker participation.

4.2 Why relationship tests are important

- If brokers simply certify deal quality, any broker might help.
- The paper finds that brokers with stronger bank concentration, longer bank relationships, and higher loyalty have larger effects.
- This is exactly the pattern predicted by repeated informal network cooperation.

Interpretation: The relationship evidence turns the paper from "brokers help" into "informal broker-bank networks help."

4.3 Alternative explanations

The paper examines three main alternatives:

- **Information/advice:** brokers simply tell clients which banks are best.
- **Certification:** brokers certify borrowers or properties to lenders.
- **Selection/liquidity:** liquidity-constrained sellers hire brokers, changing financing and price patterns.

Interpretation: The concentration and loyalty results are hard to reconcile with pure information or certification stories. The price and VTB tests suggest broker selection matters, but does not explain the bank-finance result.

4.4 Limitations

- Broker presence is not randomly assigned.
- Instruments may be debated because local broker-market structure could correlate with unobserved credit conditions.
- The paper observes completed transactions, not rejected or failed purchase attempts.
- The data do not directly observe private broker-bank communications.
- The setting is commercial real estate in the 1990s, so external validity to other lending markets must be argued by analogy.

Interpretation: The strongest evidence is the consistency of several patterns: broker presence, broker concentration, relationship longevity, loyalty, and alternative-financing tests all point toward informal networks.

5 Tables and original panels

5.1 Original prediction panels: theory implications

Read:

- Prediction 1: brokered deals are more likely to receive new bank financing.
- Predictions 2-5: brokers concentrate business among a few banks, long relationships matter, and older broker market participation raises the broker effect.
- Predictions 6-7: when bank financing is absent, brokered deals receive smaller VTB loans; brokered deals are more likely to receive assumed mortgage financing.

Economic message: The model delivers testable implications that go beyond a simple broker-selection story. The key network predictions are about repeated broker-bank relationships.

1.1 An application to commercial real estate

We will now develop implications from the theory for our commercial real estate data in order to test its implications. Here we can view the service intermediaries as brokers and the lenders as banks. We will presume that Result 1 describes the equilibrium. Good buyers who do not use brokers receive bank loans with probability q/p. Those who do not receive bank loans self-finance with probability 1/2. This implies that of nonbrokered completed transactions, a fraction $2q/(q+p)$ receive bank financing. Since banks expedite the loan applications of the clients of cooperative long-run brokers, all completed transactions of these buyers receive bank loans. The good clients of brokers who do not cooperate with a bank (whether short run or long run), receive bank financing with probability q/p. Those who do not receive bank loans self-finance with probability $(\max\{\epsilon-f,0\})/(2\epsilon)$, which implies that $(2q\epsilon)/(q(2\epsilon-\max\{\epsilon-f,0\})+p(\max\{\epsilon-f,0\}))$ of the completed transactions of these buyers receive bank finance. The inequalities

$$1 \geq \frac{2q\epsilon}{q(2\epsilon-\max\{\epsilon-f,0\})+p(\max\{\epsilon-f,0\})} > \frac{2q}{q+p} \qquad (4)$$

provide us with our first prediction.

Prediction 1. *Brokered deals are more likely to receive new bank financing.*

The clients of brokers in cooperative relationships with banks receive expedited evaluations of their loans which leads to more approvals. Clients of brokers who do not have cooperative relationships do not enjoy preferential access to loans, but they are less willing to self-finance than buyers in nonbrokered transactions, since they must pay a broker's commission in addition to engaging in costly self-financing. As a result, buyers with noncooperating brokers are also more likely to use bank finance in completed transactions than buyers without brokers.

In the equilibrium described in Result 1, cooperating brokers send all their clients to one specific bank. Cooperation is dependent upon the bank's ability to verify each period that the broker is sending it at least one buyer. We argue informally that cooperation may be possible with more than one bank, though cooperation can only be maintained with a

(a) Prediction 1

Prediction 2. *Brokers will concentrate their deals among a small number of banks.*

In the equilibrium in Result 1, there are both cooperating and noncooperating brokers. Only cooperating brokers increase their clients' probability of receiving a bank loan. Cooperating brokers will have a long history of dealing with one bank.

Prediction 3. *Brokers with long bank relationships will have a larger effect on the granting of bank loans than the average broker.*

Brokers who have dealt with a bank for multiple periods are likely to be cooperating banks who will send all their clients to the same bank in the future, as well. The empirical analog for this is that brokers and banks with long relationships should be expected to do more business with each other than with agents with whom they have short relationships.

Prediction 4. *Brokers will direct a greater proportion of their clients to banks with which they have longer relationships.*

In the equilibrium described above, short-run brokers do not develop cooperative relationships with banks. Result 2 shows that it is very hard for such cooperation to arise in equilibrium. Only long-run brokers cooperate with banks and raise their clients' loan approval rates. The formal model distinguishes only between long-run and short-run brokers, but the general intuition is that brokers who have been in the market longer are more likely to participate in cooperative relationships.

Prediction 5. *Brokers who have participated in the market for a longer period of time will have a larger effect on the granting of bank loans than the average broker.*

Buyers seeking financing may also approach a second source of loans, the seller of the property. Sellers are less diversified than banks, making loans by sellers less efficient. Such loans must offer unattractive terms to compensate sellers for the idiosyncratic risk they bear. As a result, such loans, known as vendor-to-buyer (VTB) loans, are less attractive to buyers but may be used to complete transactions.[6]

(b) Predictions 2-5

Prediction 6. *When the buyer does not receive bank financing, the size of VTB loans will be smaller in brokered deals than in nonbrokered deals.*

Finally, since banks must typically approve the assumption of an existing mortgage on a property by a new buyer, broker-bank relationships should also encourage banks to permit the assumption of old mortgages. This will only apply if the broker happens to have a relationship with the bank holding the previous mortgage on the property. It may be the case, however, that the seller will choose his broker with this criterion in mind, or that the seller will receive a broker recommendation from his bank. In both these cases, a broker-bank relationship is possible and will improve the probability that loan assumption will be permitted, in analogy with the model of new bank loans given earlier.

Prediction 7. *Brokered deals are more likely to receive assumed mortgage financing.*

Ideally we would test the conditional prediction that brokered deals are more likely to receive assumed mortgage financing when the broker and the bank have a prior relationship. It is not possible, however, to test the conditional prediction. A screen on prior bank-broker relationships will exclude all cases in which there is no new or assumed loan. (If there is no loan, then there is no bank and hence no bank-broker relationship.) This would leave us with brokers present only in transactions with new or assumed loans, and any regression would therefore exaggerate brokers' influence in aiding their clients to get assumed mortgage financing.

It is the case, however, that conditional on no bank-broker relationship, brokered deals should be equally likely to receive assumed mortgage

(c) Predictions 6-7

5.2 Table 1 + Table 2: sample structure and the main broker-finance result

Read: Table 1:

- The final sample has 22,642 commercial real estate transactions.
- Brokers are present in 65.23% of transactions.
- New mortgages appear in 52.16% of transactions; VTB finance appears in 18.80%; assumed mortgages appear in 5.32%.
- Large deals are much bigger than small deals: average sale price is about \$4.419 million versus \$355,000.

Read: Table 2:

- The first-stage instruments are jointly informative; personal crime, local cap-rate dispersion, national broker share, and broker Herfindahl all predict broker presence.
- Instrumented broker presence raises the probability of receiving a new bank mortgage; the reported coefficient is 0.757 with a t-statistic of 3.26.
- The noninstrumented broker coefficient is also positive and highly significant, but the paper emphasizes the instrumented result.
- Conditional loan size is not the main margin; broker presence is associated with a smaller new-mortgage share in the size regressions.

Economic message: Brokers mainly affect access to bank finance, not necessarily the amount of bank finance conditional on obtaining a loan.

Table 1
Summary statistics of the U.S. commercial real estate market

	All	City center	Non city	Small deals	Large deals	Apt.	Land	C&I
Panel A: Property characteristics								
No of Sales	22,642	10,815	11,827	11,325	11,317	7,924	4,134	10,584
Age[a]	35.48	42.49	28.85	39.85	31.19	37.78	0	33.63
Development	6.33%	4.74%	7.78%	5.40%	7.26%	3.50%	16.30%	4.70%
Cap. rate[a]	9.35%	9.58%	9.12%	9.16%	9.53%	10.01%	7.59%	8.26%
Personal crime rate	1.52	1.69	1.36	1.62	1.41	1.50	1.48	1.55
Property crime rate	1.92	2.37	1.50	1.99	1.85	2.04	1.60	1.94
Panel B: Market participants								
Buyer distance (median)	193.62	198.72	188.96	113.35	273.95	170.42	203.87	206.04
	(38.47)	(39.90)	(37.76)	(35.80)	(41.86)	(36.46)	(37.40)	(40.57)
Buyer out of state	11.84%	11.84%	11.84%	6.38%	17.29%	8.99%	15.43%	12.56%
Seller distance (median)	255.34	248.79	261.32	195.94	314.77	221.01	231.37	290.17
	(42.87)	(43.60)	(42.34)	(40.65)	(45.45)	(40.41)	(41.55)	(44.94)
Seller out of state	16.34%	16.26%	16.41%	12.10%	20.58%	11.75%	18.05%	19.15%
Broker present	65.23%	67.07%	63.55%	63.65%	66.82%	76.81%	46.71%	63.38%
Buyer is broker	3.22%	4.04%	2.46%	3.25%	3.18%	4.47%	3.48%	2.14%
Seller is broker	2.69%	3.26%	2.17%	2.68%	2.70%	2.73%	3.68%	2.28%
Panel C: Financial structure								
Sale price ($,000)	$2,386	$2,814	$1,995	$355	$4,419	$1,843	$1,491	$3,134
New mortgages frequency	52.16%	53.74%	50.71%	52.54%	51.77%	68.95%	25.11%	49.85%
New mortagages/Price	72.59%	70.94%	75.18%	76.57%	72.21%	75.60%	71.51%	71.16%
Vendor-to-buyer frequency	18.80%	19.47%	18.18%	23.21%	14.39%	17.66%	17.59%	20.00%
Vendor-to-buyer/Price	59.54%	60.64%	58.15%	63.27%	58.68%	50.72%	70.92%	61.17%
Assm. mortgages frequency	5.32%	5.71%	4.96%	5.03%	5.60%	9.59%	1.28%	3.65%
Assm. mortgages/Price	67.37%	69.42%	65.26%	72.23%	67.11%	70.61%	74.53%	64.10%
Loan-to-Value	72.57%	71.19%	74.62%	78.04%	71.99%	74.94%	73.60%	71.10%

Descriptive statistics on COMPS commercial real estate transactions over the period January 1, 1992, to March 30, 1999, are reported. Panel A reports general characteristics of the properties in

(a) Table 1: summary statistics

Table 2
Predicting brokerage activity and its influence on obtaining bank financing

Panel A: First-stage broker instrumented regression

Dependent variable	Population radius	Personal crime	σ_{local}	$ National/ $ Brokered	Broker Herfindahl
Broker presence (*t*-statistic)	0.037	−0.383**	−0.113*	−1.493*	−3.995**
	(0.75)	(−2.63)	(−1.79)	(−1.88)	(−2.80)

Second-stage new bank mortgage financing

Regression model:	**Panel B: Prob.(newm)** Probit		**Panel C: Size(newm/price)** Cragg (1971)	
Noncorporate buyer	0.042	0.084**	−0.003	−0.004
	(0.82)	(2.02)	(−0.73)	(−1.21)
Seller distance	−0.003**	−0.004**	0.001	0.001
	(−2.27)	(−2.06)	(1.53)	(1.35)
Buyer distance	−0.016**	−0.017**	0.001	0.001*
	(−4.02)	(−3.79)	(0.49)	(1.70)
Development	−0.039	−0.036	0.001	0.008
	(−1.02)	(−1.17)	(0.00)	(1.53)
Age	−0.001**	−0.001	0.001	0.001
	(−2.13)	(−1.49)	(−1.38)	(−0.38)
Land	−0.703**	−0.780**	−0.029	−0.016
	(−6.98)	(−7.51)	(−1.48)	(−0.86)
Apartment	0.339**	0.385**	0.029**	0.014**
	(7.92)	(7.68)	(3.15)	(3.03)
City center	0.016	0.034**	−0.001	−0.003
	(1.16)	(3.33)	(−0.20)	(−1.04)
Property crime	0.004	0.001	0.001	0.003**
	(0.42)	(−0.01)	(0.92)	(2.69)
Broker (instr.)	0.757**		−0.124*	
	(3.26)		(−1.67)	
Broker		0.380**		−0.010**
		(16.07)		(−6.40)
Log(price)	−0.017	−0.019**	−0.006**	−0.008**
	(−1.08)	(−2.51)	(−2.83)	(−6.70)

Panel A reports the first-stage regression coefficients on several exogenous instruments used to identify

(b) Table 2: broker activity and bank financing

5.3 Table 3 + Table 4: relationship concentration, longevity, herding, and loyalty

Read: Table 3:

- Brokered transactions are more concentrated among banks than nonbrokered transactions.
- BankHerf is 12.06 for brokered transactions versus 6.16 for nonbrokered transactions.
- BankShare is 21.89 for brokered transactions versus 13.60 for nonbrokered transactions.
- Broker-bank longevity positively predicts both the broker's share sent to the bank and the bank's share done by the broker.
- Broker interactions with concentration and longevity significantly raise the probability of new bank financing.

Read: Table 4:

- Broker herding into banks is positive overall and especially strong among large banks.
- Loyalty matters: loyal broker relationships increase access to bank finance, while disloyal relationships reduce it.
- The dollar-volume loyalty measures produce similar results to the transaction-count measures.

Economic message: The relationship measures are the core evidence for informal financial networks. Brokers do not just help generally; they help more when they have repeated and loyal relationships with particular lenders.

Table 3
Broker-bank relationships and their impact on capital structure

	Bank concentration measures		
	Brokered transactions	Nonbrokered transactions	Difference (*t*-statistic)
BankHerf	12.06	6.16	5.89** (3.47)
BankShare	21.89	13.60	8.29** (4.08)

	Broker-bank longevity measures		
	Dependent variable	Share of broker's deals directed to the bank	Share of the bank's deals done by the broker
Longevity of relationship in years (*t*-statistic)		0.0036** (3.15)	0.0568** (14.98)

Impact on capital structure
Probit model (Fama–Macbeth regressions)

Dependent variable	Prob.(newm)	Prob.(newm)	Prob.(newm)
Broker (instr.)	1.128** (4.31)	0.867** (4.17)	1.029** (4.41)
Broker (instr.) × concentration (*Herf*)	0.360** (10.95)		
Broker (instr.) × concentration (*share*)		0.364** (13.45)	
Broker (instr.) × broker longevity			0.020** (3.06)

(a) Table 3: broker-bank relationships

Table 4
Broker herding, loyalty, and its impact on capital structure

	Bank herding measures		
	All banks	Small banks (< 10 deals)	Large banks (≥10 deals)
Broker herding into banks	0.0222**	0.0064	0.0584**
Different from zero (*t*-statistic)	(6.08)	(1.32)	(14.89)

	Broker-bank loyalty measures		
	All brokers	Small deals (< $10 million)	Large deals (≥$10 million)
Elasticity of business	1.08	0.80**	1.38*
Different from one (*t*-statistic)	(0.77)	(−2.56)	(1.78)

Impact of loyalty on capital structure
Probit model (Fama–MacBeth regressions)

Dependent variable	Prob.(newm)	Prob.(newm)	Prob.(newm)	Prob.(newm)
Broker(instr.)	1.208** (3.33)	0.995** (4.12)	1.021** (6.06)	0.945** (3.97)
Broker(instr.) × loyalty(#)	0.266** (7.97)	0.026** (2.51)		
Broker(instr.) × disloyalty(#)		−0.019** (−2.61)		
Broker(instr.) × loyalty($)			0.214** (3.81)	0.042** (2.69)
Broker(instr.) × disloyalty($)				−0.025** (−4.35)

(b) Table 4: herding and loyalty

5.4 Table 5: alternative financing and sale prices

Read:

- Brokered deals do not show a significant instrumented effect on the unconditional probability of VTB financing.
- Conditional on no bank financing, brokered deals are more likely to have VTB financing, but the VTB loan share is smaller.
- Broker presence strongly increases the probability of assumed mortgage finance.
- Instrumented broker presence has no significant effect on cap rates, so brokered deals do not appear to receive systematically higher prices.
- Noninstrumented broker effects on prices differ from instrumented effects, showing that endogenous broker selection matters.

Economic message: Table 5 weakens the certification and liquidity-selection alternatives. Brokers appear to affect financing access through relationships rather than simply certifying asset quality or changing sale prices.

Table 5
Do brokers influence other forms of financing and the sale price?

	Panel A: Other forms of financing — Binary response probability of (Probit)				Truncated size of [Cragg (1971)]		Panel B: Sale price (Least squares)	
	VTB	VTB^a	Assm	VTB	*VTB/Price*	*VTB^a/Price*	Cap. rate	Cap. rate
Noncorp. buyer	0.152** (7.89)	0.363** (4.65)	0.120** (3.30)	0.111** (4.43)	−0.019 (−0.58)	−0.004 (−0.22)	−0.346** (−2.35)	−0.348** (−10.34)
Seller distance	0.001 (1.31)	−0.014** (−3.31)	−0.010** (−3.85)	−0.002 (−1.28)	0.003** (2.72)	0.001* (1.90)	0.011 (1.53)	0.009** (2.17)
Buyer distance	−0.018** (−8.84)	−0.056** (−3.59)	0.002 (0.70)	−0.017** (−7.47)	−0.012** (−4.12)	−0.002 (−1.52)	−0.006 (−0.87)	−0.007** (−2.46)
Development	−0.111** (−2.18)	−0.094 (−1.01)	0.012 (0.21)	−0.027 (−0.86)	0.012 (0.32)	−0.050** (−3.67)	−0.225 (−1.24)	−0.278** (−2.42)
Age	0.004** (4.58)	0.006** (2.77)	0.001 (1.53)	0.005** (4.46)	0.001** (1.98)	0.001** (2.44)	0.002** (2.25)	0.002** (1.98)
Land	−0.367** (−2.02)	−4.580* (−1.81)	−0.276** (−3.20)	−0.433** (−2.36)	−0.045** (−2.68)	−0.091** (−4.64)	−0.124 (−0.42)	−0.212 (−1.46)
Apartment	−0.089** (−2.69)	−0.025 (−0.41)	0.228** (5.16)	−0.150** (−4.31)	−0.147** (−5.50)	0.049** (4.76)	−0.299* (−1.66)	−0.246** (−3.30)
City center	0.008 (0.48)	−0.030 (−0.38)	−0.071** (−2.18)	−0.004 (−0.37)	−0.045** (−3.06)	−0.006 (−0.84)	0.029 (0.46)	−0.018 (−0.29)
Property crime	0.003 (0.42)	0.043* (1.65)	−0.003 (−0.54)	0.010** (2.03)	0.010* (1.78)	0.008** (6.44)	0.167** (4.17)	0.198** (4.55)
Broker(instr.)	−0.281 (−1.54)	1.600** (2.72)	1.433** (5.09)		−0.267* (−1.66)	−0.450** (−5.89)	0.494 (0.26)	
Broker				−0.224** (−8.30)				0.421** (6.05)
Log(price)	−0.165**	−0.268**	0.081**	−0.190**	−0.036**	−0.015**	−0.144	−0.089**

Figure 4: Table 5: other financing forms and sale prices

6 Short summary

- The paper studies informal financial networks in U.S. commercial real estate.
- The theoretical model shows how nonfinancial intermediaries can increase borrowers' access to bank finance.
- Brokers repeatedly send clients to banks; banks reciprocate by expediting loan evaluations.
- This repeated relationship is valuable because individual buyers transact infrequently and cannot credibly promise future business.
- The empirical sample contains 22,642 commercial real estate transactions from 1992-1999.
- Brokered deals are more likely to obtain new bank mortgage financing after instrumenting for broker presence.
- Brokers concentrate business among a small number of banks.
- Brokers with more concentrated and longer bank relationships have larger effects on bank finance.
- Loyal brokers increase the probability of bank financing; disloyal brokers reduce it.
- Broker presence increases assumed mortgage financing and changes VTB patterns in ways consistent with the model.
- Brokered deals do not receive higher prices after instrumentation, weakening a simple certification interpretation.
- The central lesson is that informal noncontractual relationships can shape capital access even in mature U.S. financial markets.

7 Conclusion

7.1 Core conclusion

Garmaise and Moskowitz argue that financial access is partly mediated by informal networks. Service intermediaries such as brokers can influence lenders because they repeatedly channel future business. The empirical evidence from commercial real estate shows that brokered transactions are more likely to receive bank financing, especially when brokers have concentrated, loyal, and long-standing relationships with banks.

7.2 Economic intuition

A bank may not find it worthwhile to expedite a one-time borrower's application. But a broker can reward the bank with future client referrals. This repeated-game logic transforms a costly one-shot action into a sustainable relationship. The broker benefits because more transactions close and generate fees; the borrower benefits because bank finance becomes more accessible; and the bank benefits from future deal flow.

7.3 Takeaway

The paper expands the theory of financial intermediation beyond formal lenders. It shows that informal networks can allocate credit by linking borrowers to banks through repeated nonfinancial intermediaries. The implication is broad: credit access may depend not only on borrower fundamentals and bank balance sheets, but also on relationship capital embedded in local service networks.